PROJECT PROPOSAL

Smart Air Quality and Pollution Analytics

Enrollment Numbers: 250A52041, 250A52050, 250A52060
Batch: 02

1. Introduction

Smart Air Quality and Pollution Analytics is a data analytics and visualization project designed to bring air quality information from monitoring stations across India onto a single, user-friendly platform. It helps users understand pollution through AQI monitoring, pollutant analysis, interactive maps, historical trends, and city-wise comparisons.

2. Problem Statement

Air quality data is published in raw and scattered formats and is often difficult for ordinary users to interpret. People may struggle to identify polluted areas, understand which pollutants are responsible, or see whether air quality is improving over time.

This project addresses the problem by converting raw monitoring data into a centralized analytics dashboard with clear visualizations, pollution analysis, and station-level information.

3. Aim and Objectives

Aim:

To develop an interactive dashboard that analyzes and visualizes air quality and pollutant data across India to provide useful environmental insights.

Objectives:

- To display the current AQI and AQI category for cities and stations.
- To monitor pollutant levels such as PM2.5, PM10, NO_2, SO_2, CO, and O_3.
- To visualize AQI trends and compare air quality across cities and states.
- To identify the most polluted cities and pollution hotspots using an interactive map.
- To show the distribution of days across AQI categories.

4. Target Users

- **General Public:** To understand daily air quality and pollution levels.
- **Health-Sensitive Individuals:** To know when air quality is unsafe during poor pollution days.
- **Environmental Authorities:** To monitor pollution patterns and high-risk areas.

5. Proposed Methodology

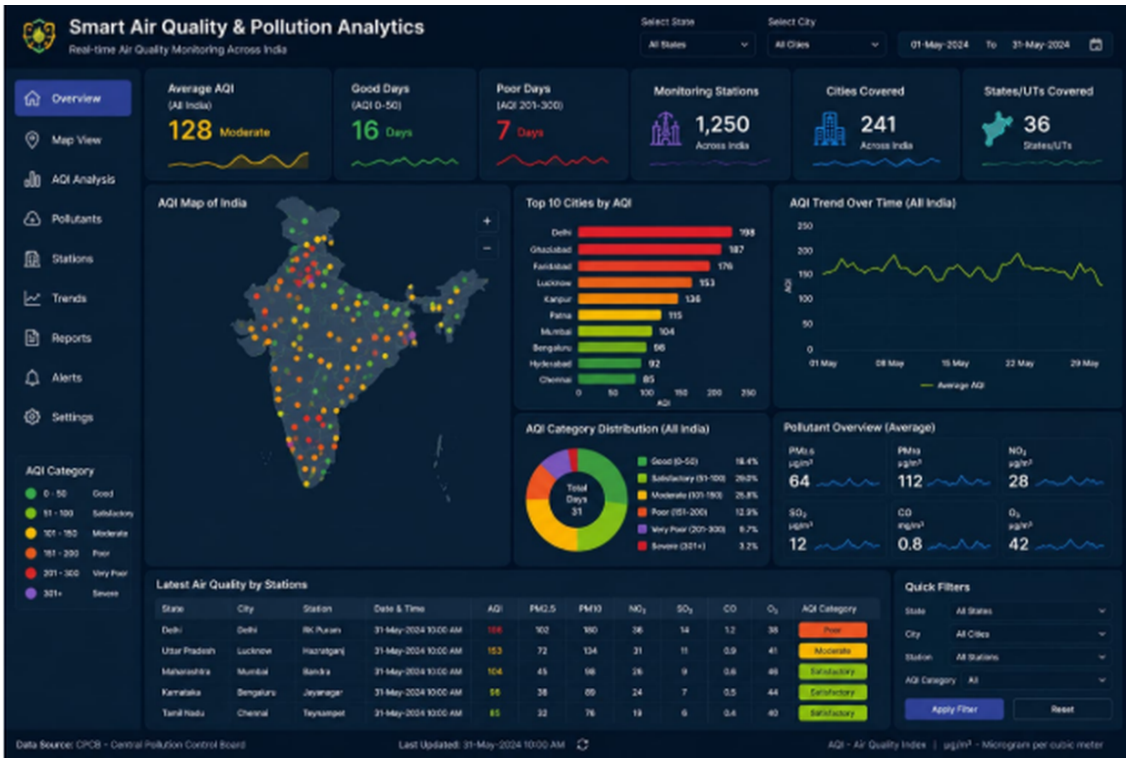
The project will be developed through the following steps:

1. **Data Collection:** Collect air quality data from suitable APIs and reliable sources such as CPCB.
2. **Data Cleaning:** Remove or handle missing, duplicate, and incorrect data to improve data quality.
3. **Data Analysis:** Analyze AQI, pollutant levels, and historical and city-wise trends.
4. **Data Visualization:** Represent the collected data using charts, graphs, and interactive maps.
5. **Interactive Dashboard:** Develop a user-friendly dashboard to display AQI levels, pollutant overview, pollution hotspots, and station-wise data.

6. Proposed Features

- AQI monitoring with AQI category distribution.
- Pollutant monitoring for PM2.5, PM10, NO_2, O_3, CO, and SO_2, where data is available.
- Charts showing historical AQI trends and city-wise comparisons.
- Interactive maps showing pollution hotspots.
- Station-wise air quality table with state, city, station, and date filters.

7. Expected Dashboard



8. Conclusion

Smart Air Quality and Pollution Analytics aims to make air quality information easier to access and understand. By combining data analysis, interactive maps, and clear visualizations, the project will support environmental awareness and informed decisions.